

Logarithms



1

a $\log_5 25 = 2$

c $\log_3 3 = 1$

e $\log_{4.4} 1 = 0$

g $\log_4 32 = \frac{5}{2}$

i $\log_4 \frac{1}{64} = -3$

b $\log_3 81 = x$

d $\log_2 1 = 0$

f $\log_{25} 125 = 1.5$

h $\log_4 0.0625 = -2$

j $\log_x 64 = 1.5$

2

a $\log_{10} 100 = 2$

c $\log_{10} (\sqrt{10}) = \frac{1}{2}$

b $\log_{10} \left(\frac{1}{100} \right) = -2$

d $\log_{10} \left(\frac{1}{\sqrt{10}} \right) = -\frac{1}{2}$

3

a $4^2 = 16$

e $3^{-1} = \frac{1}{3}$

i $8^6 = x$

b $5^1 = 5$

f $(5.8)^2 = 33.64$

c $8^0 = 1$

g $\left(\frac{1}{3} \right)^{-2} = 9$

d $2^{-3} = 0.125$

h $x^5 = 32$

4

a $10^4 = 10\,000$

b $10^{\frac{1}{2}} = \sqrt{10}$

c $10^{-4} = \frac{1}{10\,000}$

d $10^{-\frac{1}{2}} = \frac{1}{\sqrt{10}}$

5 True

6 1

7

a 4

e $\frac{1}{2}$

i 3

m $-\frac{4}{3}$

b -2

f -2

j $\frac{1}{4}$

n $\frac{15}{4}$

c -1

g -1

k 1

d 0

h $\frac{1}{3}$

l $\frac{1}{8}$

8 0.713

9

a 1

c Between 1 and 2



b 2

10

a Between 3 and 4

b 3.9490

11

a No

b No

c Yes, 0.30

d Yes, 1.26

12 3

13

a 6

b 0

c 8

d -3

14 True

15 $x = 25$

16 1.32

17 $k = 4$

18

a 4

b $x = 1$

19

a

i $x = \log_{10} 790$

ii 2.90

b

i $x = \log_{10} 0.0003$

ii -3.52

20

a -1

b 9

c -2

d $\frac{1}{2}$

e $-\frac{1}{2}$